DIGITAL TWIN AI · PROTOTYPE

IBD Digital Twin

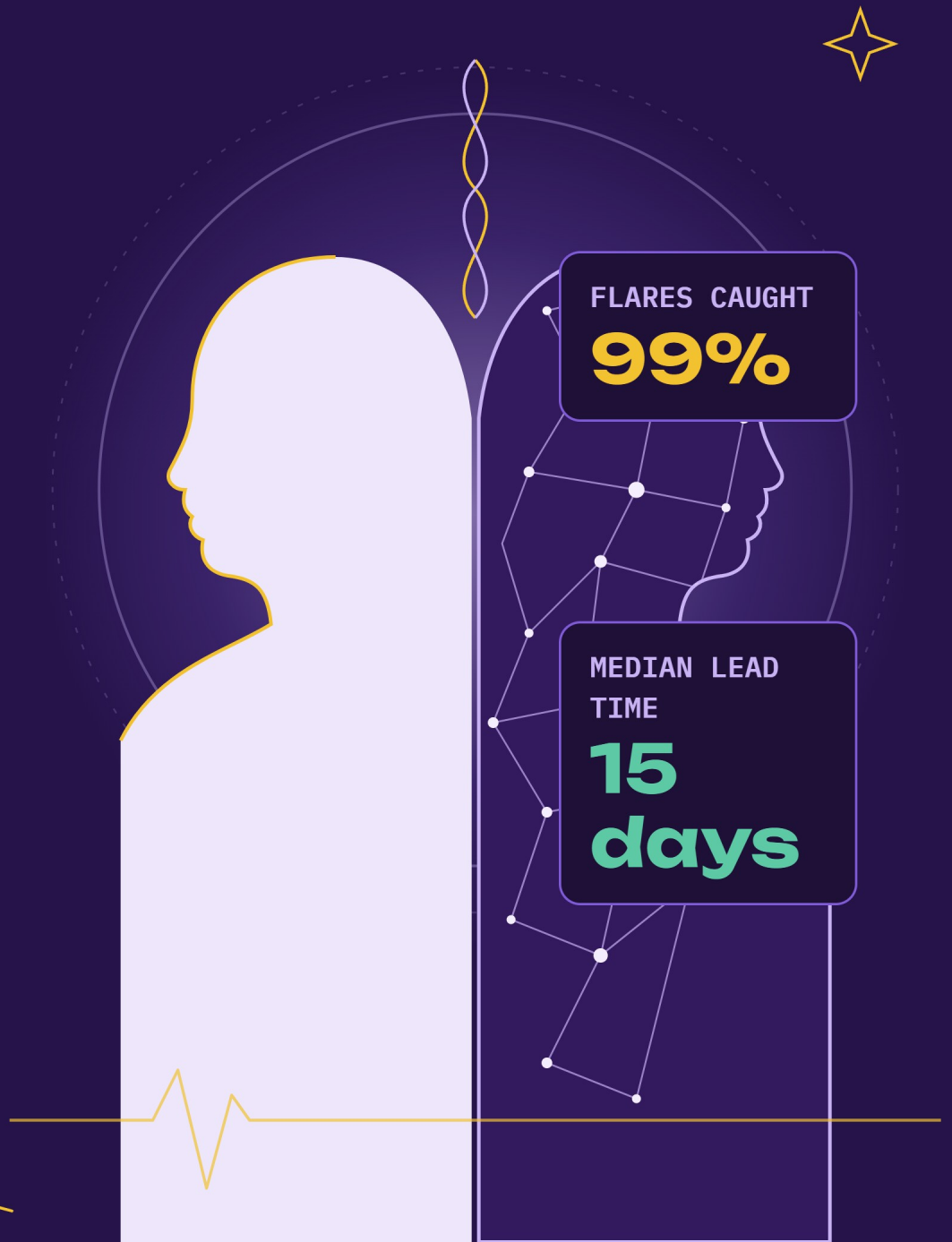
Predicting inflammatory bowel disease flares weeks before they hit

14-DAY FLARE RISK | 28-DAY OUTLOOK | WHAT-IF SIMULATOR

Presented by **Prantik Sarkar** and **Swapnil Ghatak**

prantik.pgp26 · IIM Shillong

Synthetic data only · flags risk, never diagnoses



India has Asia's largest IBD population, and it is growing fast

~1.4M

patients in India (2010 est.), largest in Asia

and rising

20–40

peak age: hits the working years



15 mo

median diagnostic delay



5.5×

monthly household income lost to that delay (₹1.36 lakh)



Acceleration stage

India is where Korea and Japan were 20–30 years ago (Nature 2025).

Not just a city disease

No rural–urban difference among 30,000+ symptomatic patients (AIG), but specialists sit in tier-1 cities.

No cure

Care means keeping patients out of flares, yet today we react after the flare has hit.

The body warns us weeks before a flare

EARLY-WARNING TIMELINE · WEEKS BEFORE FLARE ONSET



wearables

7 weeks

Smartwatch HRV falls, resting HR rises and steps drop up to 7 weeks before IBD flares; F1 up to 0.90

HIRTEN ET AL., MOUNT SINAI, GASTROENTEROLOGY 2025

lab test

3 months

Stool calprotectin > 300 three months before UC relapse; two consecutive highs → 100% specificity

ANTWERP INFLIXIMAB COHORT

symptoms

Mismatch

Symptoms lag inflammation and mislead; wearables + labs resolve the symptom–inflammation mismatch

SYSTEMATIC REVIEW, REPEATED CALPROTECTIN

A virtual patient that mirrors, predicts, simulates and protects

01 Mirror

EHR + watch + daily check-in + labs, each signal against the patient's own baseline

EHR
WATCH
CHECK-IN
LABS



02 Predict

Calibrated risk of a flare starting in 14 days, plus a 28-day outlook

TRAFFIC LIGHT

- **Low** < 20%
- **Rising** 20–50%
- **High** ≥ 50%

03 Simulate

What-if scenarios re-score the twin (model-estimated)

ADHERENCE 100%
SWITCH TO ANTI-TNF
STOP NSAIDS

04 Protect

Emergency rules fire within hours; a Cancer Safety panel tracks colonoscopy surveillance

6

EMERGENCY RULES

+ CANCER SAFETY

Five layers, one-way data flow, ground truth kept out

1 Synthetic patients and data streams `src/generator/`
800 virtual Indian IBD patients × 365 days (seed 42); latent inflammation drives every stream

EHR · Wearables · Check-in · Labs · Sweat (inv.)

2 Fusion and personal baselines `src/features/`
One row per patient-day from data ≤ day t, each signal vs the patient's own quiet-day baseline

z-scores · 7/14/28-day · Lab deltas · Context flags

3 Prediction, explanation and what-if `src/models/`
Logistic vs LightGBM, 5-fold CV grouped by patient, Platt-calibrated

14-day risk · TreeSHAP top-5 · What-if · Ablation A–E

4 Alerts and safety rules `src/alerts/`
Trend alerts over days to weeks; emergency patterns within hours

≥20% → calprotectin · ≥50% → teleconsult · 6 emergency rules

5 Doctor dashboard `src/dashboard/app.py`
Population view by risk; patient card, gauges, why, what-if, alerts; a date slider replays any day

Streamlit · Cancer Safety

Ground truth

never a model input

Events, latent inflammation and hidden parameters build the flare-within-14-days label and the evaluation only.

Design choices

- ✓ Layers 2–5 disease-agnostic (COPD, heart failure next)
- ✓ Sensor-agnostic: new streams are new columns
- ✓ Out-of-fold scores in the UI

STACK: PYTHON 3.11 · PANDAS · NUMPY · PYARROW · SCIKIT-LEARN · LIGHTGBM · STREAMLIT · PLOTLY · PYTEST

800 synthetic patients × 365 days, built to be hard

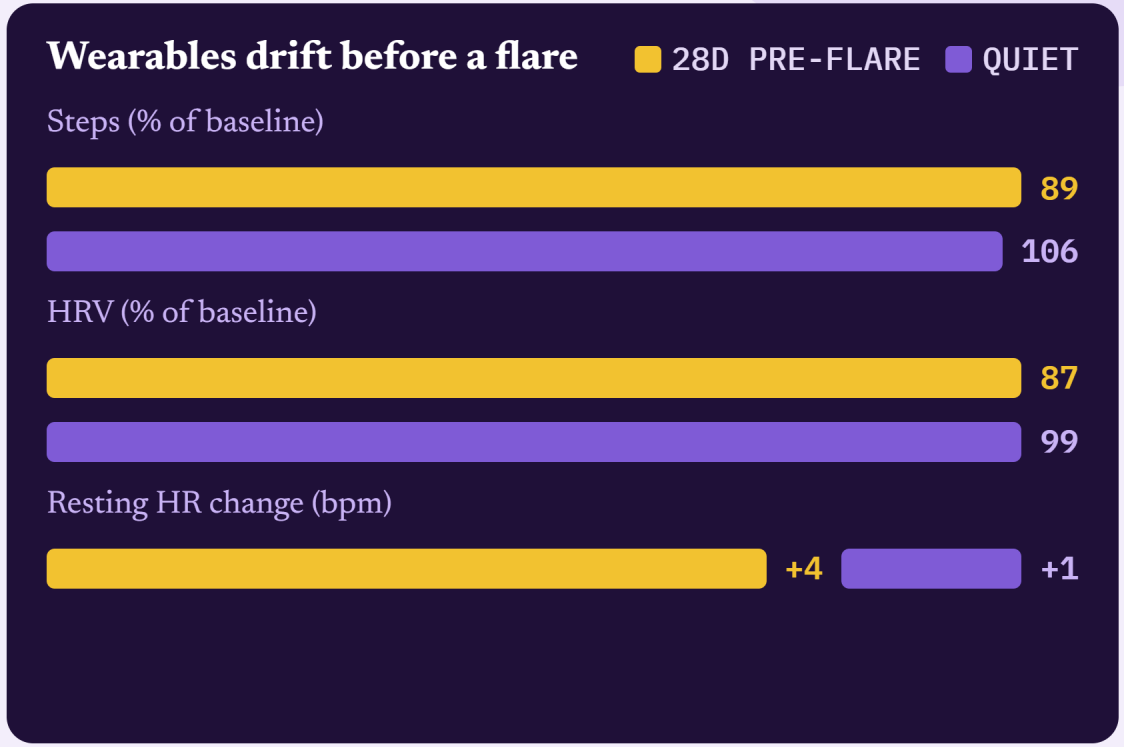
- ✓ Latent inflammation drives every stream, with lags and noise
- ✓ 8–12 week pre-flare ramp; 20% of episodes settle on their own
- ✓ "Silent" patients and IBS overlap (symptoms without inflammation)
- ✓ 17 seeded demo patients, incl. Ananya and 6 emergency scenarios

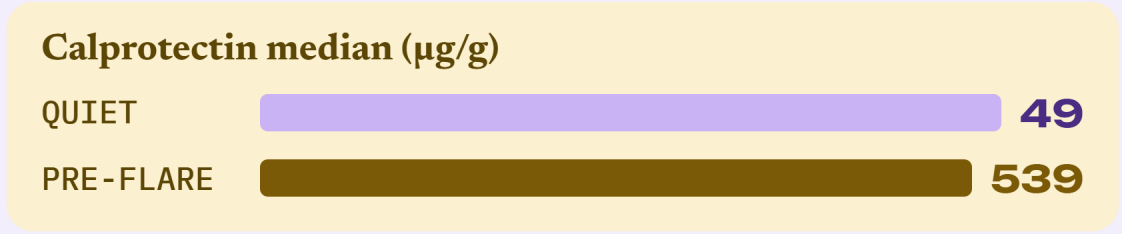
Decoys that fool a naive model

on purpose

1,562	2,234	565	286
COLDS	STRESS WKS	NSAID COURSES	GUT INFECTIONS

WATCH WORN	83%	CHECK-INS DONE	73%





Personal baselines, causal features, patient-grouped validation

Personal baseline 01



Median / MAD of each signal over the patient's own quiet days (no cold, NSAID, steroid or high calprotectin)

Features at day t 02



7 · 14 · 28 DAYS → t

z-scores; mean, SD, min, max, slope; co-movement; lab deltas; symptom trends; context flags

Model 03

LOGISTIC vs LIGHTGBM



Class weights; Platt calibration; threshold set on validation patients

Validation 04



5 FOLDS · BY PATIENT

Out-of-fold scores in the dashboard; a leakage test scrambles all data after t

TARGET

A flare onset (inflammatory and symptomatic) in days **t+1 to t+14**

2.3% POSITIVE ROWS — 97.7% NEGATIVE



so we report PR-AUC

Catches 99% of flares about 2 weeks ahead and rarely cries wolf

0.881
PR-AUC on held-out patients
LOGISTIC 0.844

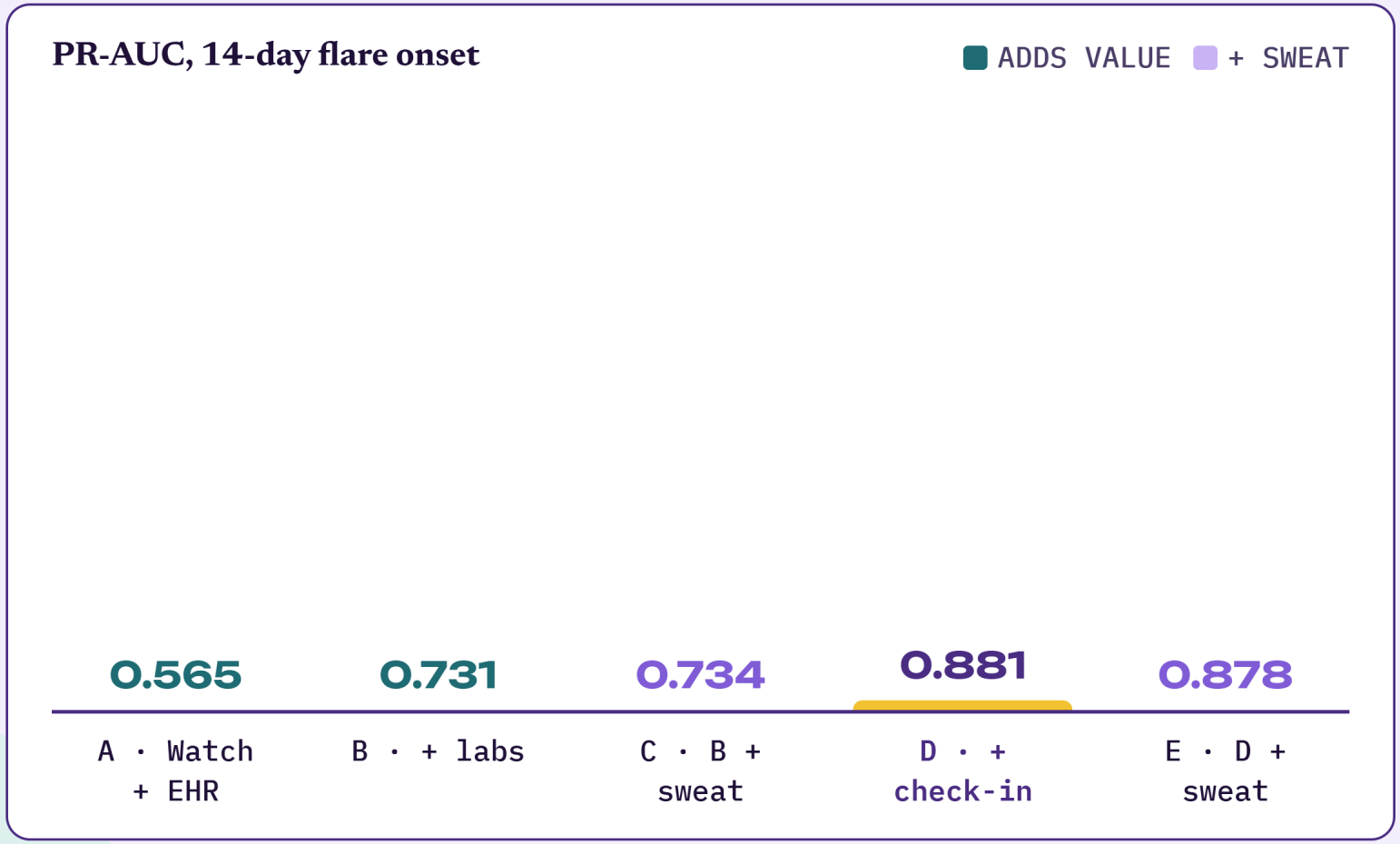
99%
of flare onsets caught (408 of 411) by an alert in the 14 days before

15 days
median warning lead time
28-DAY OUTLOOK: 20 DAYS

0.038
false alarms per patient-month
≈ 1 PER 26 MONTHS

Held-out patients	LightGBM	Logistic
AUROC	0.995	0.990
F1 at threshold	0.810	0.792
Sensitivity / specificity (days)	0.86 / 0.99	0.81 / 0.99
Brier score	0.0063	0.0072

Every stream earns its place, except the sweat sensor (for now)



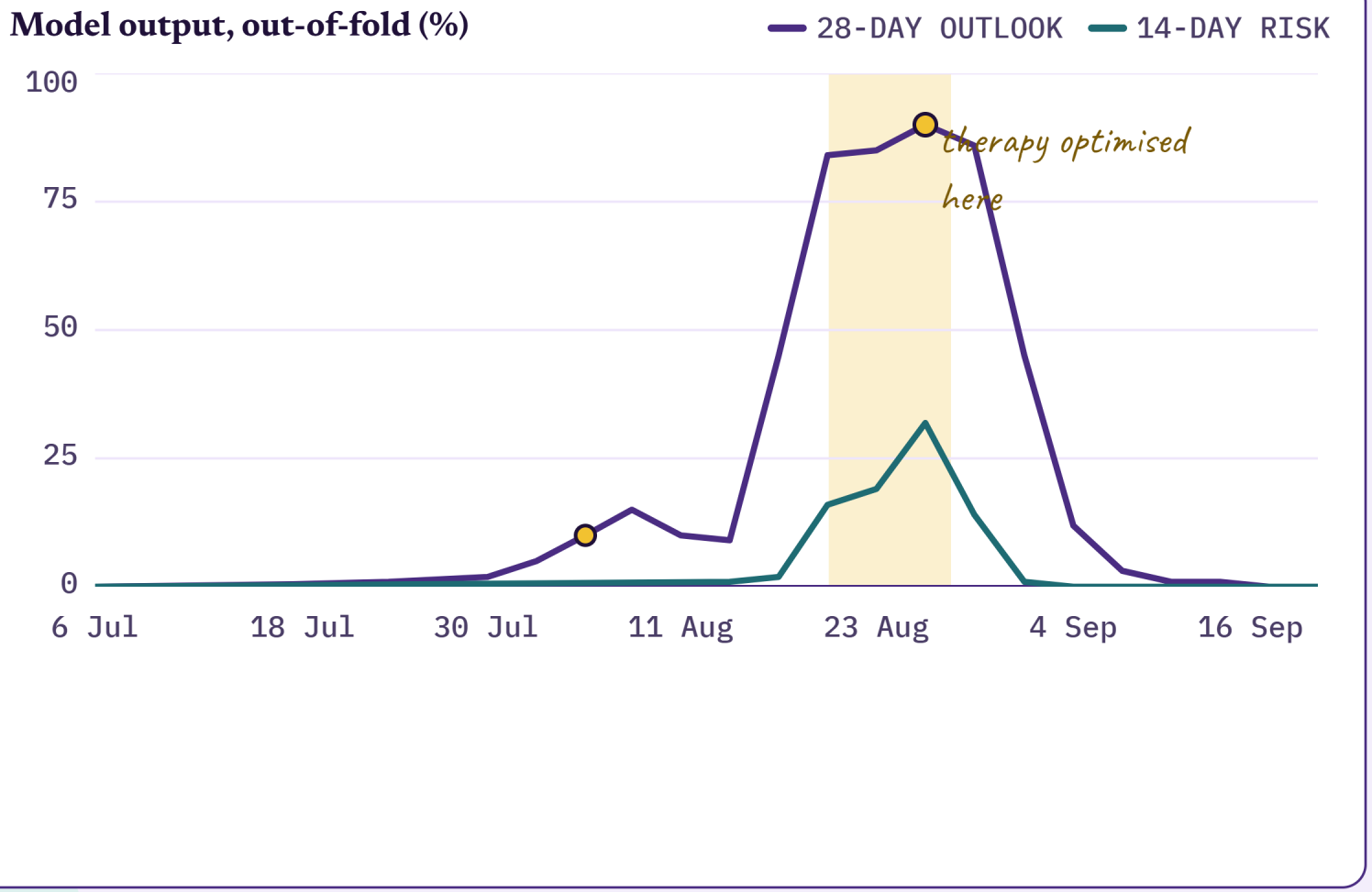
Watch alone finds 68%
of flares passively, with no patient effort.

Calprotectin adds most per test
PR-AUC 0.57 → 0.73; 90% of flares.

A 30-second check-in closes the gap
99% of flares; symptoms rise in the final 2 weeks.

Sweat sensor: +0.003
Stays investigational until it beats stool tests.

Ananya, 28, Bengaluru: a flare averted



WK 1 · 13 JUL
Low: at baseline

WK 4 · 11 AUG
Rising: watch drift on 7 of 14 days, no cold → home calprotectin

WK 5 · 17–18 AUG
High: calprotectin 310, outlook 53%; HRV rhythm −45%, RHR 64→69

WK 6 · 19–24 AUG
Therapy optimised; outlook peaks while signals settle

WK 9 · 14 SEP
Low: back to baseline, flare averted

The doctor's view: status, why, what-if and next step

Ananya R. · P0001
28 y F · urban, Karnataka · ulcerative colitis · 5-ASA

TWIN STATUS: HIGH

14-DAY RISK
3%

28-DAY OUTLOOK
53%

SUGGESTED ACTION
Teleconsult this week and review therapy

WHY IS THE RISK AT THIS LEVEL?

- HRV rhythm −45%
- Resting HR +5.4 bpm
- Calprotectin 310 µg/g
- CRP 3.0 mg/L
- Resp. rate +0.7/min

WHAT-IF SIMULATOR

- Adherence = 100%
- Switch to anti-TNF
- Stop NSAIDs
- Optimise (all three)

Population view
All patients sorted by emergency, then status; overdue colonoscopies

Patient view
Card, 14- and 28-day gauges, signals vs baseline band, labs, alerts

Why + what-if
Top-5 SHAP drivers in plain words; re-score with therapy changes

Replay
A date slider sets "today" for any patient

Emergencies in hours, cancer surveillance in years

WITHIN HOURS Six emergency rules

Emergency rule	Pattern	Seeded
Acute severe colitis	≥6 bloody stools + RHR > 90 + fever	P0002–04
VTE / PE	RHR jump + SpO2 drop + immobility	P0005–06
Sepsis on steroids	Fever + RHR + resp. rate ↑ + HRV crash	P0007–08
Bowel obstruction	Vomiting + no stool/gas, stricture	P0009–10
Stoma dehydration	High output + RHR ↑ + dizziness	P0011–12
TB reactivation	Fevers + night sweats + weight loss, anti-TNF	P0016–17

Every rule fires on its seeded patients and on < **0.005%** of other patient-days

Cancer Safety

YEARS

- ✓ CRC risk level (BSG 2019 + inflammation burden)
- ✓ Next colonoscopy in 1 / 3 / 5 years; PSC yearly; overdue flag

1 Y | 3 Y | 5 Y

- ⚠ Red flags: falling Hb, weight loss, bleeding in remission, labs worse while watch is stable
- ✓ Skin, HPV, PSC and TB screening tracker

surveillance support, not cancer detection

Start with the watch people already own

TODAY — NEXT — LATER

ready now

Today

Smartwatch + home calprotectin + 30-second check-in; teleconsult for rural patients far from a specialist

WATCH · HOME TEST · TELECONSULT

pilot

Next

Pilot with a gastroenterology centre under ethics approval and DPDP Act consent; CDSCO software-as-medical-device pathway

ETHICS · DPDP · CDSCO SaMD

scale up

Later

Continuous sweat-biomarker sensors once validated (CDSCO investigational route); extend the twin to COPD and heart failure

SWEAT SENSOR · COPD · HEART FAILURE

What this prototype does not prove

read before you trust the numbers

Synthetic data

Correlations exist because the generator puts them there; drift sizes are literature-informed assumptions

Not causal

What-if re-scores the model; therapy is a static EHR field, so projected changes are small and not treatment effects

Non-specific signals

Colds, stress and alcohol move the watch too; mitigated by logged decoys, labs and symptoms

Confounded labs

Calprotectin rises with NSAIDs, infections, PPIs, TB and CRC; CRP is not gut-specific

Investigational sensor

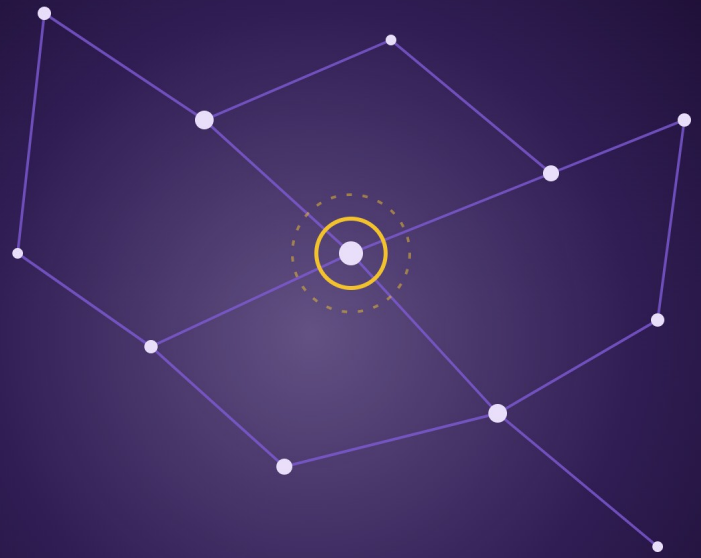
No FDA or CDSCO approval for sweat-based IBD monitoring

Not validated in India

Needs ethics approval, DPDP compliance, a CDSCO pathway and clinician oversight

IBD DIGITAL TWIN

A first step toward catching IBD flares earlier



- Flares caught weeks ahead
- Emergencies within hours
- No missed colonoscopy

Prantik Sarkar · Swapnil Ghatak · IIM Shillong



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